

**Boston Women Entrepreneurship Incubator  
Tracker**

**Milestone: Conceptual Models EER and UML**

Group 2

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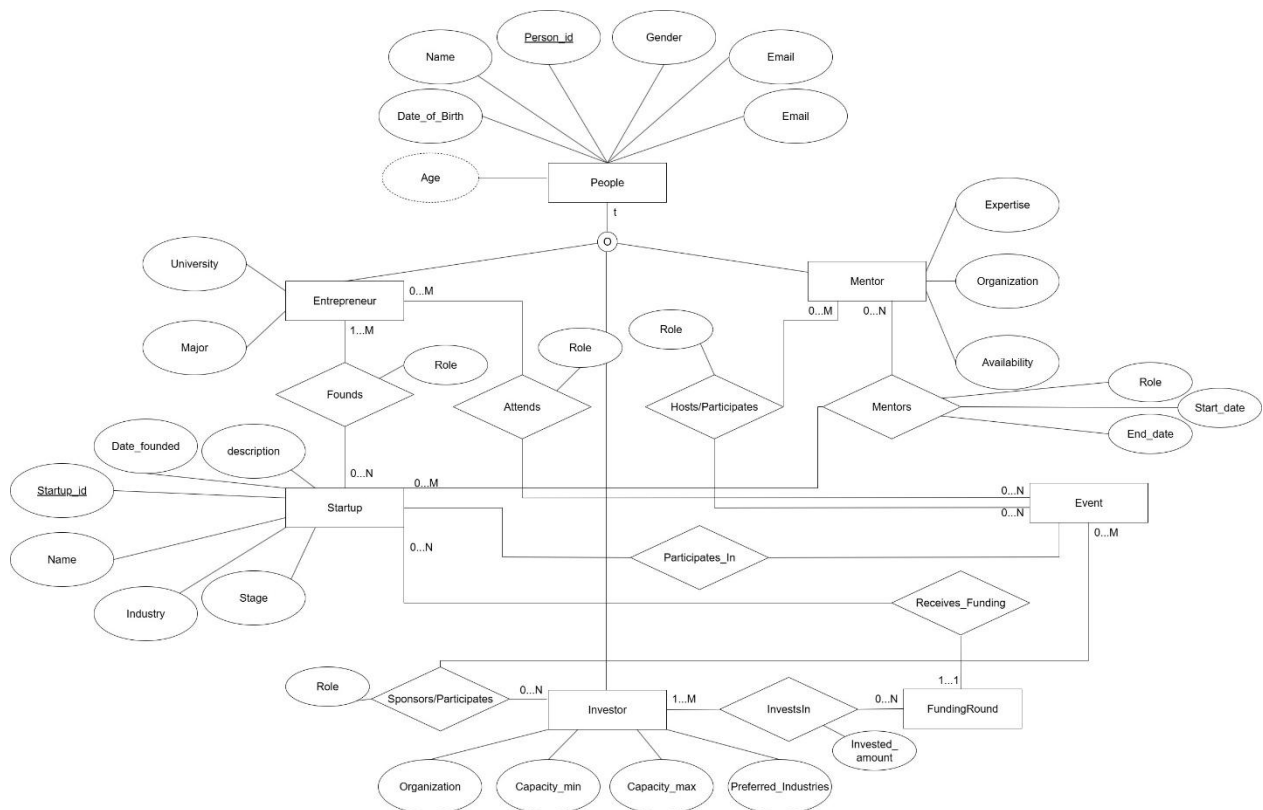
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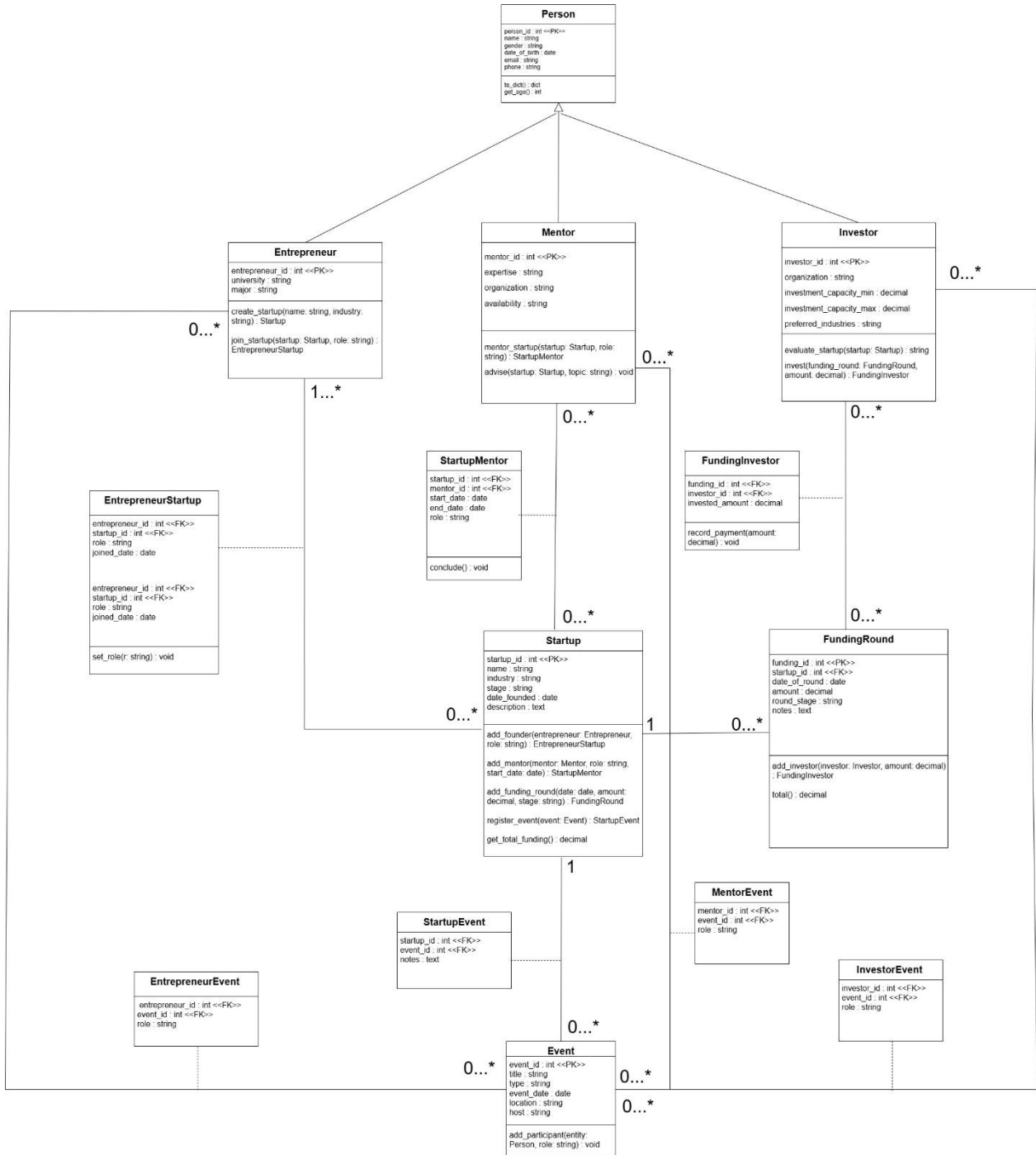
# Boston Women Entrepreneurship Incubator Tracker

## Rutuja Dambir and Praniti Kale

### 1. EER DIAGRAM



## 2. UML DIAGRAM



## Domain Information

The Boston Women Entrepreneurship Incubator Tracker is a platform that functions as part of Boston's innovation ecosystem by connecting women student entrepreneurs with mentors, investors, and University opportunities.

The incubator is designed to help student founders develop their startups, receive mentorship, attract investment, and engage in their entrepreneurial journey through workshops, pitch competitions, and networking.

This system captures and organizes all of this information through both:

An Enhanced Entity-Relationship (EER), which provides a conceptual view of structure, entity attributes, and cardinality, as well as

A Unified Modeling Language (UML) that captures dynamic information such as interactions, class hierarchies, and relation across processes.

The combined models are a strong foundation for operations management and reporting on the performance of women-led ventures across industry and university.

## Business Rules That Cannot Be Fully Modelled

Although the EER and UML frameworks represent the majority of operational logic, there are some real-world constraints that cannot be specified directly in a diagram or modeling structure. These will require application-level logic, validation scripts, or policy enforcement to ensure constraints are adhered to.

| AREA              | Business Rule (Unmodeled Constraint)                                                                                                                                |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Entrepreneur      | Only current students or recent graduates may register as entrepreneurs.<br>Each entrepreneur may found a maximum of three startups simultaneously.                 |
| Mentor            | A mentor can guide no more than five startups at once.<br>Mentor-startup pairing should align by industry expertise                                                 |
| Investor          | Investments must remain within each investor's declared funding capacity range.<br>Investors cannot fund directly competing startups within the same funding round. |
| Startup & Funding | Each funding round belongs to one startup but can include multiple investors.                                                                                       |

|               |                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event         | Participant roles must align with their type (e.g., only investors may act as sponsors).<br>Each event may host a maximum of 100 participants due to logistical constraints. |
| Data & Access | Records are retained for five years post-program; all personal data must comply with FERPA and GDPR.                                                                         |

**EER Model View**

The EER model defines the objects and how they can relate to one another and cardinalities (for example, one startup can have many mentors or funding rounds). However, for example the EER cannot say “no more than five startups can have the same mentor”, since numerical or time restrictions cannot be modeled as: now about operational and ethical policies, a process or topics which further extends the logic of design beyond just the schema.

**UML Model View**

Similarly, associations, generalizations and use cases in UML class diagrams and interaction diagrams, demonstrate how objects interact within the incubator system but do not fully document human or policy-based constraints, (starting from eligibility, role levels, time eligibility, etc.) which also depend on business and procedural processes comprising further decision-making operations from outside the system.